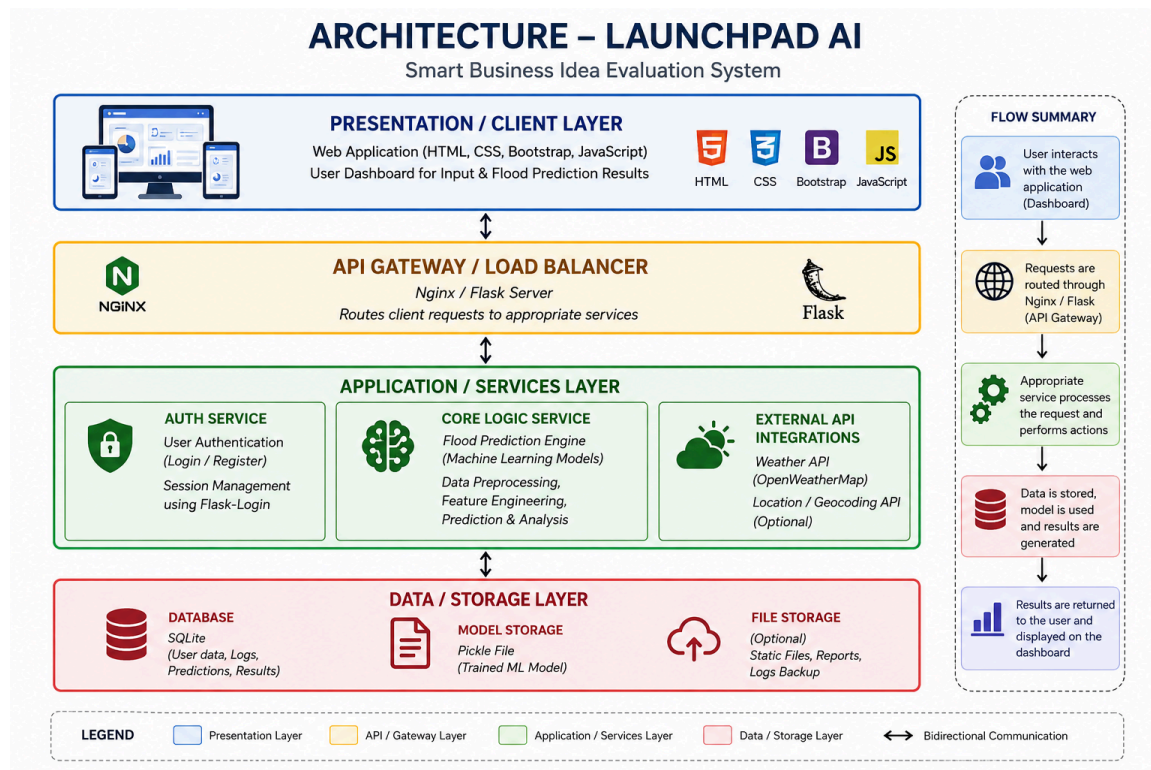


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	LaunchPad AI – Smart Business Idea Evaluation System
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a responsive web interface where users enter startup details and view AI-generated startup analysis reports.	HTML, CSS, JavaScript, Flask Templates
API Gateway/Application Layer	Receives user requests, validates inputs, manages routes, communicates with Gemini AI, and returns analysis results.	Flask (Python)
Authentication Module	Manages user sessions and controls access to application features (if enabled).	Flask Session Management
Core Logic Service	Builds AI prompts, sends requests to Google Gemini AI, processes responses, and generates structured startup analysis, SWOT analysis, market analysis, and recommendations.	Python, Google Gemini AI API, Markdown, Pydantic
External AI Integration	Connects the application with Google Gemini AI to generate intelligent business insights based on user input.	Google Gemini AI API
Service Modules	Implements the core flood prediction service, including data preprocessing, feature scaling, loading the trained XGBoost model, and generating flood predictions.	Flask Blueprints, Python
Database	Stores startup information, generated AI reports, and analysis history for future reference.	SQLite, SQLAlchemy
Deployment Layer	Hosts the Flask application online and manages production deployment with automatic updates from GitHub.	Render, Gunicorn, GitHub